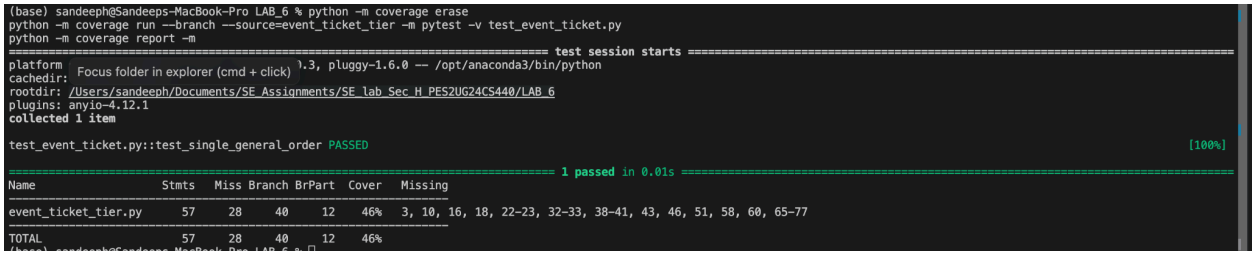
Lab 6: Code and Branch Coverage Analysis

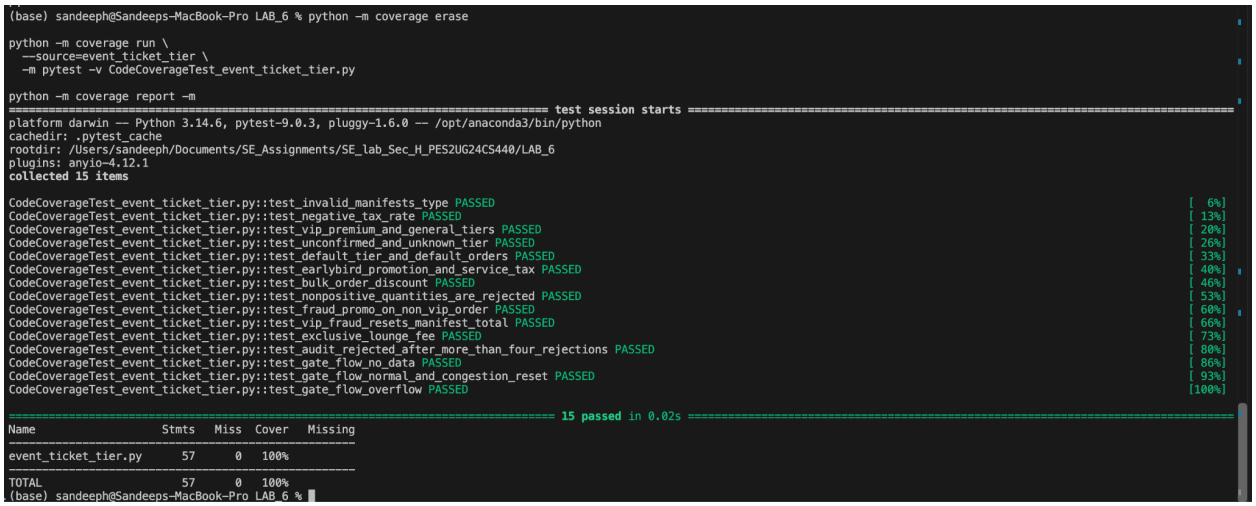
NAME:Sandeep H SRN:PES2UG24CS440

Before:

Branch Coverage:

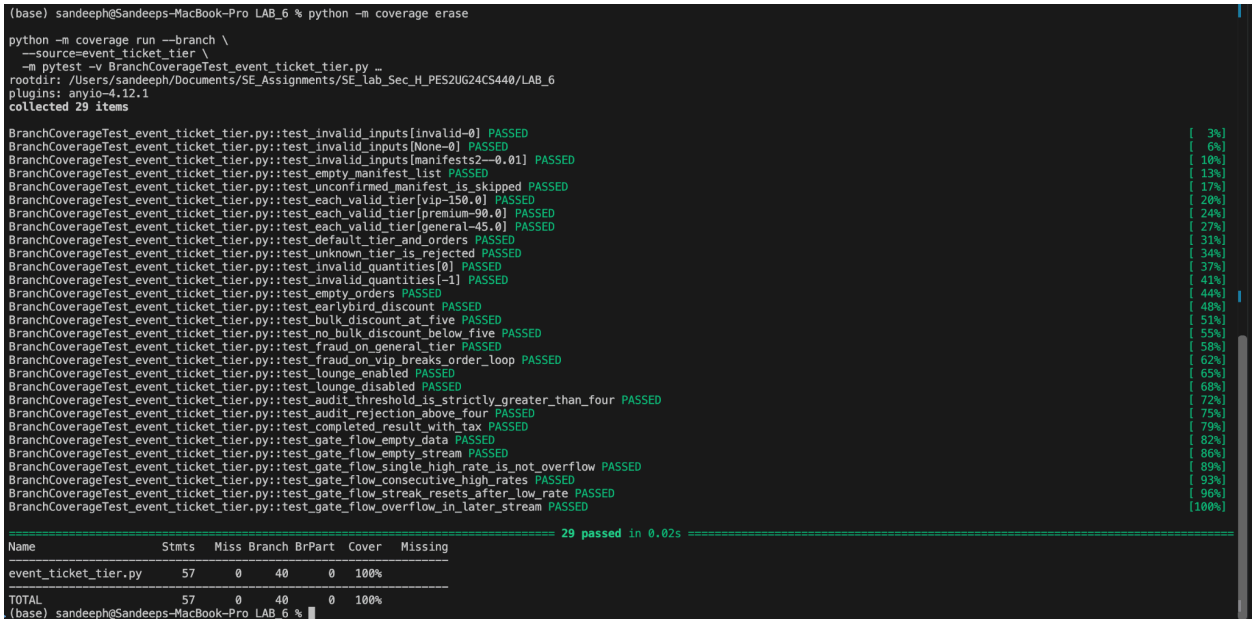


Line Coverage



After:

Branch Coverage:



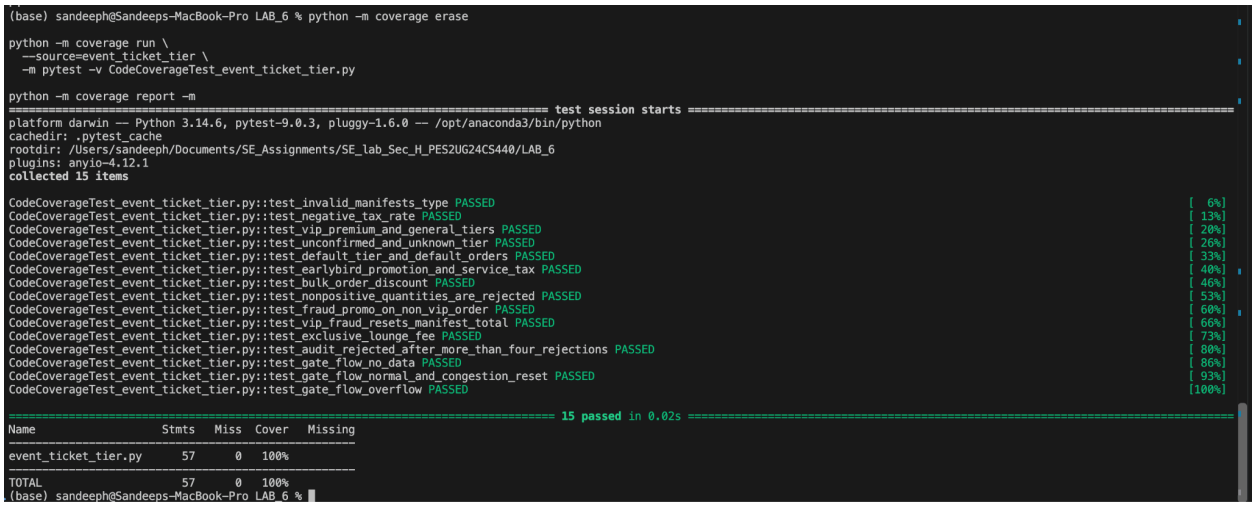
```
(base) sandeeph@Sandeeps-MacBook-Pro LAB_6 % python -m coverage erase

python -m coverage run --branch \
  --source=event_ticket_tier \
  -m pytest -v BranchCoverageTest_event_ticket_tier.py …
rootdir: /Users/sandeeph/Documents/SE_Assignments/SE_lab_Sec_H_PES2UG24CS440/LAB_6
plugins: anyio-4.12.1
collected 29 items

BranchCoverageTest_event_ticket_tier.py::test_invalid_inputs[invalid-0] PASSED                                                                [  3%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_inputs[None-0] PASSED                                                                   [  6%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_inputs[manifests2--0.01] PASSED                                                         [ 10%]
BranchCoverageTest_event_ticket_tier.py::test_empty_manifest_list PASSED                                                                      [ 13%]
BranchCoverageTest_event_ticket_tier.py::test_unconfirmed_manifest_is_skipped PASSED                                                          [ 17%]
BranchCoverageTest_event_ticket_tier.py::test_each_valid_tier[vip-150.0] PASSED                                                               [ 20%]
BranchCoverageTest_event_ticket_tier.py::test_each_valid_tier[premium-90.0] PASSED                                                            [ 24%]
BranchCoverageTest_event_ticket_tier.py::test_each_valid_tier[general-45.0] PASSED                                                            [ 27%]
BranchCoverageTest_event_ticket_tier.py::test_default_tier_and_orders PASSED                                                                  [ 31%]
BranchCoverageTest_event_ticket_tier.py::test_unknown_tier_is_rejected PASSED                                                                 [ 34%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_quantities[0] PASSED                                                                    [ 37%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_quantities[-1] PASSED                                                                   [ 41%]
BranchCoverageTest_event_ticket_tier.py::test_empty_orders PASSED                                                                             [ 44%]
BranchCoverageTest_event_ticket_tier.py::test_earlybird_discount PASSED                                                                       [ 48%]
BranchCoverageTest_event_ticket_tier.py::test_bulk_discount_at_five PASSED                                                                    [ 51%]
BranchCoverageTest_event_ticket_tier.py::test_no_bulk_discount_below_five PASSED                                                              [ 55%]
BranchCoverageTest_event_ticket_tier.py::test_fraud_on_general_tier PASSED                                                                    [ 58%]
BranchCoverageTest_event_ticket_tier.py::test_fraud_on_vip_breaks_order_loop PASSED                                                           [ 62%]
BranchCoverageTest_event_ticket_tier.py::test_lounge_enabled PASSED                                                                           [ 65%]
BranchCoverageTest_event_ticket_tier.py::test_lounge_disabled PASSED                                                                          [ 68%]
BranchCoverageTest_event_ticket_tier.py::test_audit_threshold_is_strictly_greater_than_four PASSED                                            [ 72%]
BranchCoverageTest_event_ticket_tier.py::test_audit_rejection_above_four PASSED                                                               [ 75%]
BranchCoverageTest_event_ticket_tier.py::test_completed_result_with_tax PASSED                                                                [ 79%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_empty_data PASSED                                                                     [ 82%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_empty_stream PASSED                                                                   [ 86%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_single_high_rate_is_not_overflow PASSED                                               [ 89%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_consecutive_high_rates PASSED                                                         [ 93%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_streak_resets_after_low_rate PASSED                                                   [ 96%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_overflow_in_later_stream PASSED                                                       [100%]

============================================================================== 29 passed in 0.02s ==============================================================================
Name                    Stmts   Miss Branch BrPart  Cover   Missing
-------------------------------------------------------------------------
event_ticket_tier.py       57      0     40      0   100%
-------------------------------------------------------------------------
TOTAL                      57      0     40      0   100%
(base) sandeeph@Sandeeps-MacBook-Pro LAB_6 %
```

Line Coverage



```
(base) sandeeph@Sandeeps-MacBook-Pro LAB_6 % python -m coverage erase

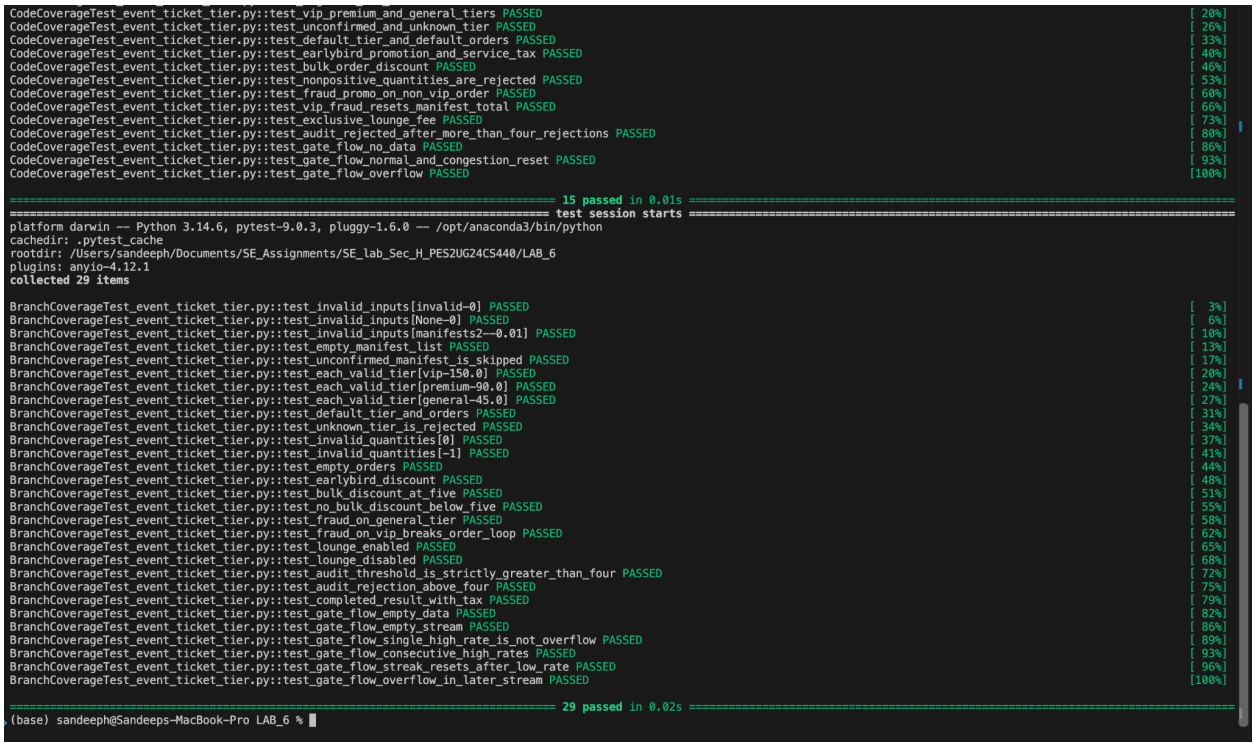
python -m coverage run \
  --source=event_ticket_tier \
  -m pytest -v CodeCoverageTest_event_ticket_tier.py

python -m coverage report -m
============================================================================== test session starts ==============================================================================
platform darwin -- Python 3.14.6, pytest-9.0.3, pluggy-1.6.0 -- /opt/anaconda3/bin/python
cachedir: .pytest_cache
rootdir: /Users/sandeeph/Documents/SE_Assignments/SE_lab_Sec_H_PES2UG24CS440/LAB_6
plugins: anyio-4.12.1
collected 15 items

CodeCoverageTest_event_ticket_tier.py::test_invalid_manifests_type PASSED                                                                     [  6%]
CodeCoverageTest_event_ticket_tier.py::test_negative_tax_rate PASSED                                                                          [ 13%]
CodeCoverageTest_event_ticket_tier.py::test_vip_premium_and_general_tiers PASSED                                                              [ 20%]
CodeCoverageTest_event_ticket_tier.py::test_unconfirmed_and_unknown_tier PASSED                                                               [ 26%]
CodeCoverageTest_event_ticket_tier.py::test_default_tier_and_default_orders PASSED                                                            [ 33%]
CodeCoverageTest_event_ticket_tier.py::test_earlybird_promotion_and_service_tax PASSED                                                        [ 40%]
CodeCoverageTest_event_ticket_tier.py::test_bulk_order_discount PASSED                                                                        [ 46%]
CodeCoverageTest_event_ticket_tier.py::test_nonpositive_quantities_are_rejected PASSED                                                        [ 53%]
CodeCoverageTest_event_ticket_tier.py::test_fraud_promo_on_non_vip_order PASSED                                                               [ 60%]
CodeCoverageTest_event_ticket_tier.py::test_vip_fraud_resets_manifest_total PASSED                                                            [ 66%]
CodeCoverageTest_event_ticket_tier.py::test_exclusive_lounge_fee PASSED                                                                       [ 73%]
CodeCoverageTest_event_ticket_tier.py::test_audit_rejected_after_more_than_four_rejections PASSED                                             [ 80%]
CodeCoverageTest_event_ticket_tier.py::test_gate_flow_no_data PASSED                                                                          [ 86%]
CodeCoverageTest_event_ticket_tier.py::test_gate_flow_normal_and_congestion_reset PASSED                                                      [ 93%]
CodeCoverageTest_event_ticket_tier.py::test_gate_flow_overflow PASSED                                                                         [100%]

============================================================================== 15 passed in 0.02s ==============================================================================
Name                    Stmts   Miss  Cover   Missing
-----------------------------------------------------
event_ticket_tier.py       57      0   100%
-----------------------------------------------------
TOTAL                      57      0   100%
(base) sandeeph@Sandeeps-MacBook-Pro LAB_6 %
```

Final Pytest :



```
CodeCoverageTest_event_ticket_tier.py::test_vip_premium_and_general_tiers PASSED                                    [ 20%]
CodeCoverageTest_event_ticket_tier.py::test_unconfirmed_and_unknown_tier PASSED                                     [ 26%]
CodeCoverageTest_event_ticket_tier.py::test_default_tier_and_default_orders PASSED                                  [ 33%]
CodeCoverageTest_event_ticket_tier.py::test_earlybird_promotion_and_service_tax PASSED                              [ 40%]
CodeCoverageTest_event_ticket_tier.py::test_bulk_order_discount PASSED                                              [ 46%]
CodeCoverageTest_event_ticket_tier.py::test_nonpositive_quantities_are_rejected PASSED                              [ 53%]
CodeCoverageTest_event_ticket_tier.py::test_fraud_promo_on_non_vip_order PASSED                                     [ 60%]
CodeCoverageTest_event_ticket_tier.py::test_vip_fraud_resets_manifest_total PASSED                                  [ 66%]
CodeCoverageTest_event_ticket_tier.py::test_exclusive_lounge_fee PASSED                                             [ 73%]
CodeCoverageTest_event_ticket_tier.py::test_audit_rejected_after_more_than_four_rejections PASSED                   [ 80%]
CodeCoverageTest_event_ticket_tier.py::test_gate_flow_no_data PASSED                                                [ 86%]
CodeCoverageTest_event_ticket_tier.py::test_gate_flow_normal_and_congestion_reset PASSED                            [ 93%]
CodeCoverageTest_event_ticket_tier.py::test_gate_flow_overflow PASSED                                               [100%]

============================================================================== 15 passed in 0.01s ==============================================================================
============================================================================== test session starts ==============================================================================
platform darwin -- Python 3.14.6, pytest-9.0.3, pluggy-1.6.0 -- /opt/anaconda3/bin/python
cachedir: .pytest_cache
rootdir: /Users/sandeeph/Documents/SE_Assignments/SE_lab_Sec_H_PES2UG24CS440/LAB_6
plugins: anyio-4.12.1
collected 29 items

BranchCoverageTest_event_ticket_tier.py::test_invalid_inputs[invalid-0] PASSED                                      [  3%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_inputs[None-0] PASSED                                         [  6%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_inputs[manifests2--0.01] PASSED                               [ 10%]
BranchCoverageTest_event_ticket_tier.py::test_empty_manifest_list PASSED                                            [ 13%]
BranchCoverageTest_event_ticket_tier.py::test_unconfirmed_manifest_is_skipped PASSED                                [ 17%]
BranchCoverageTest_event_ticket_tier.py::test_each_valid_tier[vip-150.0] PASSED                                     [ 20%]
BranchCoverageTest_event_ticket_tier.py::test_each_valid_tier[premium-90.0] PASSED                                  [ 24%]
BranchCoverageTest_event_ticket_tier.py::test_each_valid_tier[general-45.0] PASSED                                  [ 27%]
BranchCoverageTest_event_ticket_tier.py::test_default_tier_and_orders PASSED                                        [ 31%]
BranchCoverageTest_event_ticket_tier.py::test_unknown_tier_is_rejected PASSED                                       [ 34%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_quantities[0] PASSED                                          [ 37%]
BranchCoverageTest_event_ticket_tier.py::test_invalid_quantities[-1] PASSED                                         [ 41%]
BranchCoverageTest_event_ticket_tier.py::test_empty_orders PASSED                                                   [ 44%]
BranchCoverageTest_event_ticket_tier.py::test_earlybird_discount PASSED                                             [ 48%]
BranchCoverageTest_event_ticket_tier.py::test_bulk_discount_at_five PASSED                                          [ 51%]
BranchCoverageTest_event_ticket_tier.py::test_no_bulk_discount_below_five PASSED                                    [ 55%]
BranchCoverageTest_event_ticket_tier.py::test_fraud_on_general_tier PASSED                                          [ 58%]
BranchCoverageTest_event_ticket_tier.py::test_fraud_on_vip_breaks_order_loop PASSED                                 [ 62%]
BranchCoverageTest_event_ticket_tier.py::test_lounge_enabled PASSED                                                 [ 65%]
BranchCoverageTest_event_ticket_tier.py::test_lounge_disabled PASSED                                                [ 68%]
BranchCoverageTest_event_ticket_tier.py::test_audit_threshold_is_strictly_greater_than_four PASSED                  [ 72%]
BranchCoverageTest_event_ticket_tier.py::test_audit_rejection_above_four PASSED                                     [ 75%]
BranchCoverageTest_event_ticket_tier.py::test_completed_result_with_tax PASSED                                      [ 79%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_empty_data PASSED                                           [ 82%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_empty_stream PASSED                                         [ 86%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_single_high_rate_is_not_overflow PASSED                     [ 89%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_consecutive_high_rates PASSED                               [ 93%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_streak_resets_after_low_rate PASSED                         [ 96%]
BranchCoverageTest_event_ticket_tier.py::test_gate_flow_overflow_in_later_stream PASSED                             [100%]

============================================================================== 29 passed in 0.02s ==============================================================================
(base) sandeeph@Sandeeps-MacBook-Pro LAB_6 %
```